FINAL REPORT / OCTOBER 10, 2026

How Fast Does Normal Happen?

Stress-Testing the Central Limit Theorem with Monte Carlo Simulation

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The answer is population-specific. I investigated how large one sample must be for its average to have a useful Normal approximation. The practical choices range from exact Normality at n = 1 to a qualified dependent-process choice at 350. Cauchy has no finite answer.

Report at a Glance

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All three extra-credit investigations are included: dependent machine failure, non-identical basketball/ Bernoulli shots, and the synthetic age-at-death mixture. Their cards are labeled **Extra credit**; the assumption discussion and validation provide additional evidence.

Scope: five required populations and three extra-credit processes, with B = 10,000 fresh Monte Carlo samples per tested n. Results are practical approximation judgments, not universal thresholds.

GitHub: babaJaan01/Monte_Carlo_CLT

02 / Three questions, three answers

Research question: What is the smallest tested n at which I am comfortable calling the sampling distribution of the sample mean approximately Normal?

Question 1 / How large must n be?

There is no universal n=30. Thirty exceeds the Normal, die and age choices, but clearer departures remain for Exponential, small Binomial, basketball and dependent failure. It cannot repair Cauchy.

Fastest: Normal is exact at 1. The symmetric bounded die (10) and bounded synthetic mixture (15) are fast non-Normal iid cases. An irregular population is not necessarily slow.

Slower: Exponential (90) retains right skew; Binomial (75) and basketball (140) retain discrete gaps. Basketball has small skew at its choice, yet steps matter. Dependent failure (350) is the slowest finite empirical choice, with extra uncertainty about asymptotic behavior.

I base this conclusion on the master results (page 5), the eight evidence cards (pages 6-9), and smaller-n comparisons (page 10).

Question 2 / What counts as approximately Normal?

A bell-shaped histogram alone was not enough. Q-Q curvature/steps, skewness, excess kurtosis and nearby stability supported my practical choices, not a single p-value. No numerical pass/fail cutoff was used. These choices do not certify extreme-tail accuracy.

I base this conclusion on the explicit decision standard and Normal calibration (page 4), plus the graphical and numerical boundary evidence (pages 6-10).

Question 3 / What if CLT assumptions fail?

Cauchy’s tails invalidate the ordinary finite-variance theorem. Dependence retains covariance terms in Var(Xbar). Independent non-identical shots can still have a useful approximation with their individual variances. Symmetry, skew, discreteness, boundedness and dependence mattered here; no one property explained every result.

I base this conclusion on the extra-credit and Cauchy cards (pages 8-9), the special-case interpretation (page 11), and appropriate mean/SE checks (page 12).

Limit: the ranking depends on this tolerance and tested grid. Stricter tail accuracy could require larger n. More B improves simulation precision; it does not repair a violated assumption. Borderline neighboring choices may also be defensible.

03 / AI helped; iteration improved the argument

The initial prompt asked AI for a professional reusable R/Quarto project with eight cases, B = 10,000, calibration and no invented conclusions. I requested clear decisions and evidence-backed reasons rather than complicated output.

Stage	AI contribution / check	My actual input
Build	Generators, simulation, diagnostics, Quarto	Specified scope, R, reproducibility and understandable code
Technical audit	Found overwrite/shared-grid defects; fixed and tested them	Requested careful correction; did not personally find the code bugs
Boundary reasoning	Smaller candidates, plots and two-seed comparisons	Questioned prettier plots; approved practical tolerance
Final report	Compressed 35 pages; checked results and layout	Required a concise submission with clear answers and evidence

A real exchange, not an invented iteration

My follow-up: “They looked better. But I didn’t know … we need the smallest defensible n. Maybe then going off visually isn’t right?”

Weakness: nicer larger-n plots did not explain why smaller values were unacceptable. Demanding exact-Normal simulation noise was too strict for a practical approximation.

Revision: targeted smaller-n checks, below/at/above comparisons, fixed alternate seed and explicit borderline reservations. Die 9 versus 10 now acknowledges ambiguity rather than a manufactured sharp boundary.

Validation: both seeds’ saved vectors reproduce their rows; seeded replay preserves the machine recurrence; tests verify varying basketball probabilities, their SE and age moments. These are computational/theoretical checks, not a claim that AI is an independent reviewer.

Understanding the important methods

One generator call creates n observations; one mean produces one number; B repetitions create the sampling distribution. Histograms show central shape, Q-Q bends show distributional mismatch, and correct mean/SE does not establish shape. Cauchy’s population moments are undefined even when a finite sample average is calculable. These distinctions were clarified during the recorded review; complete line-by-line mastery is not claimed.

Decision ownership: AI proposed the working integers under the practical standard I approved. My contribution was specifying the task, questioning the smallest-n argument and requesting corrections and validation. The log separates those roles; it does not invent personal predictions, reactions or independent code audits.

The real prompts and corrections remain in `ai/ai_interactions.md`; important-code explanations are in the separate investigations, not full listings here.

04 / What counts as normal enough?

My standard is a **useful central-shape approximation**, not exact Normality or certified accuracy for rare tail probabilities. Small residual bends and discrete steps may be acceptable. No single diagnostic, hypothesis test or p-value decides the result. The project used **no numerical pass/fail cutoffs**.

Evidence	What I looked for	Role in the decision
Histogram	A broadly symmetric, unimodal bell	Visual support, not enough alone
Normal Q-Q plot	No substantial systematic curvature	Separate tail bends from lattice steps
Skewness / excess kurtosis	Modest asymmetry and tail discrepancy	Describe departures; no rigid thresholds
Q-Q RMSE	Smaller standardized rank discrepancies	Compare candidates and Normal noise
Mean / SE	Agreement with valid theoretical references	Check implementation; spread is not shape
Stability	Similar conclusions nearby and under seed 7617	Avoid one unusually nice plot

Monte Carlo design

At each n I generate **B = 10,000 fresh samples**, average the n observations in each, then diagnose the B means. This is not bootstrapping an observed dataset. The original population does not become Normal; its averages may.

The 15-value coarse grid spans n=1-500. Population-specific refinements and large-n checks produced **215 tested case/n combinations per seed**. Smaller candidates and nearby larger values informed the choices; the detailed files retain the progression.

Seed **7616** starts the fixed-order primary experiment; **7617** starts the identical alternate design. Targeted boundary experiments reset their assigned seed before each experiment. Seed and experiment order both matter.

Calibration: a reference, not a cutoff

Thirty genuinely Normal experiments at each coarse n (each B = 10,000) quantify simulation noise. At n = 1 their empirical 2.5th-97.5th percentile ranges were:

Diagnostic	Normal-only noise range
Skewness	-0.033 to 0.032
Excess kurtosis	-0.070 to 0.106
Q-Q RMSE	0.010 to 0.018

These are empirical ranges, not exact tolerance limits; accepted cases can remain outside this noise under the practical standard.

Reading the plots: histograms show sample means; orange Normal overlays use valid theoretical mean/SE only. Q-Q lines are fitted quartile references. Discrete bins respect the 1/n lattice. QQ RMSE compares sorted, empirically standardized means with standard-Normal rank quantiles; it is descriptive, not proof.

Expectations: no pre-run personal predictions were recorded. The card expectations are retrospective theory.

05 / The decisions at a glance

Numerical entries come from the executed row at the selected n, seed 7616, B = 10,000. Kurt. means **excess** kurtosis; QQ means standardized Q-Q RMSE. N/A is not zero.

Population	Shape	Tested n (count)	n	Skew	Kurt.	QQ	Key observation
Normal	Exactly Normal	1-500 (15)	1	0.003	-0.022	0.011	Exact at every n
Six-sided die	Symmetric, discrete	1-500 (32)	10	0.000	-0.146	0.057	Fine steps; short tails
Exponential	Right tail	1-1000 (28)	90	0.214	0.032	0.053	Right skew remains
Binomial small	Rare discrete counts	1-1000 (29)	75	0.109	0.050	0.057	Steps may remain
Cauchy	Infinite moments	1-5000 (26)	None	N/A	N/A	N/A	Averages stay Cauchy
Dependent failure*	Dependent recurrence	1-1000 (29)	350	0.208	0.097	0.051	Empirical; no iid SE
Basketball shots*	Independent, varying p	1-1000 (29)	140	-0.052	-0.039	0.057	Use individual variances
Synthetic ages*	Bounded mixture	1-500 (27)	15	-0.191	-0.039	0.050	Irregular shape smooths

*** Extra credit:** all three marked cases are included. Normal is exact; the other finite choices are practical, with dependent failure qualified. Cauchy has no finite choice by theory.

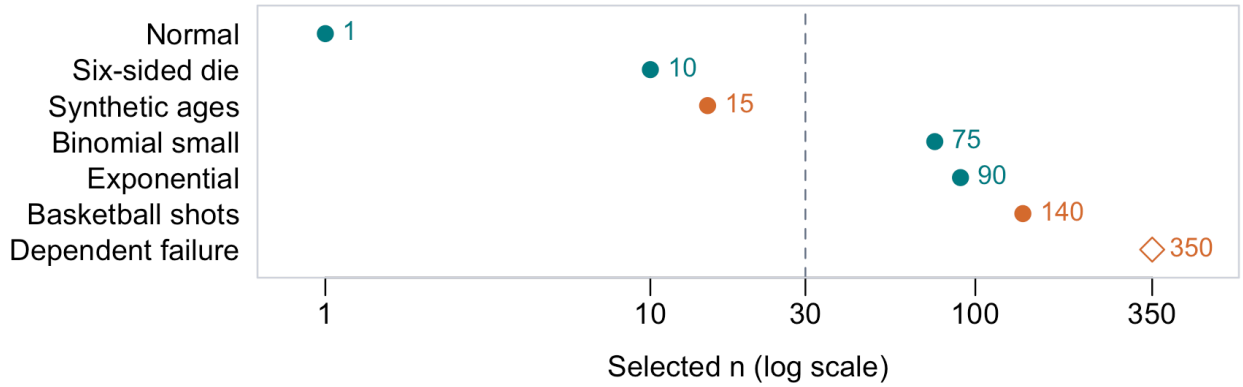


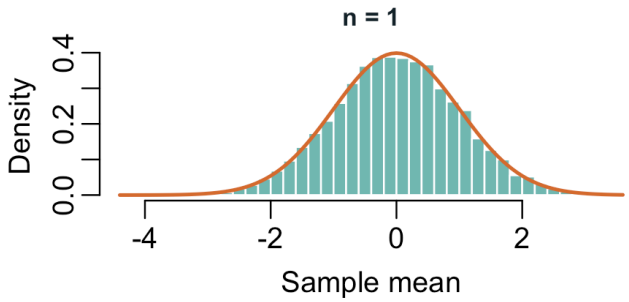
Figure 1: Selected n on a log scale. Teal: required; orange: extra credit; open diamond: qualified dependent-process choice. Dashed n=30 is not an acceptance threshold. No finite Cauchy point exists.

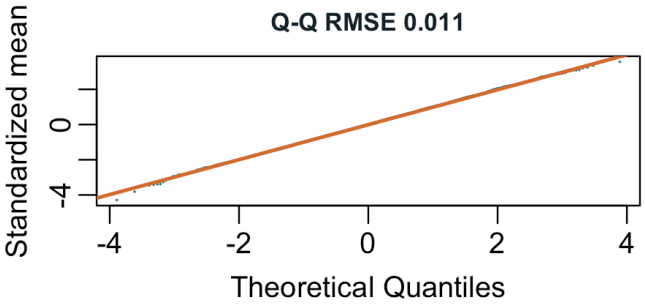
06 / Exact and fast

Normal n = 1

iid N(0,1); mean 0, SD 1. Theory expectation: exact Normality at n=1.

Tested: 1-500 (15 values); neighborhood 1, 2.





Metrics: skew 0.003; excess kurtosis –0.022; QQ 0.011.

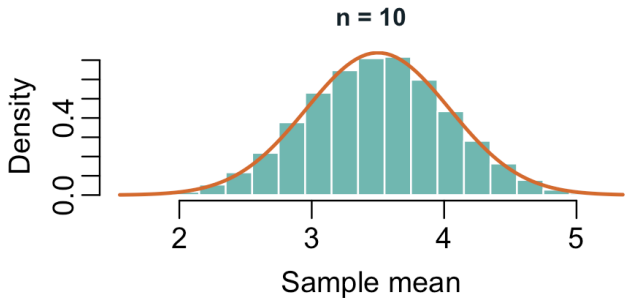
Why this n? The bell, Q-Q line and SE agree with exact Normal theory. Small nonzero diagnostics are simulation noise, not a reason to increase n.

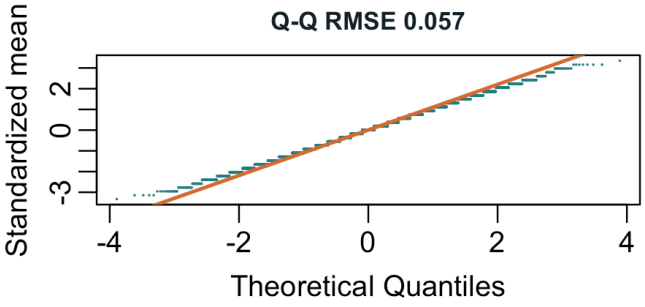
Observation: Normality is exact; averaging only narrows spread.

Six-sided die n = 10

iid outcomes 1-6; mean 3.5, SD sqrt(35/12). Theory expectation: fast smoothing with steps.

Tested: 1-500 (32 values); neighborhood 9, 10, 12.





Metrics: skew 0.000; excess kurtosis –0.146; QQ 0.057.

Why this n? The symmetric bell and broadly linear Q-Q plot support 10 despite fine steps and short-tail error. Both seeds agree; 6-9 are borderline, not proved failures.

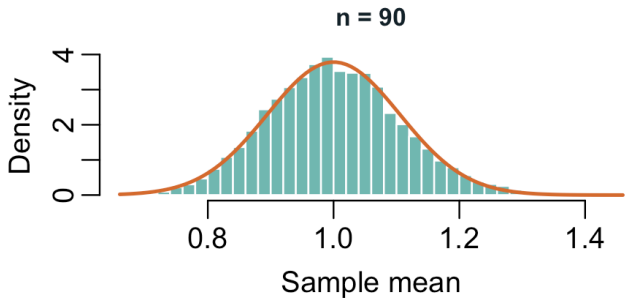
Observation: Approximately Normal does not mean step-free.

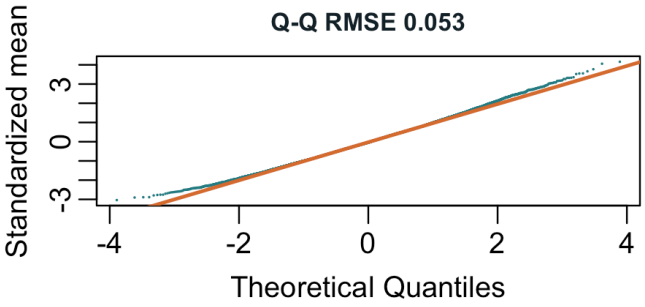
07 / Skew and rare counts

Exponential n = 90

iid Exponential(rate=1); mean and SD 1. Theory expectation: right skew fades gradually.

Tested: 1-1000 (28 values); neighborhood 80, 90, 100.





Metrics: skew 0.214; excess kurtosis 0.032; QQ 0.053.

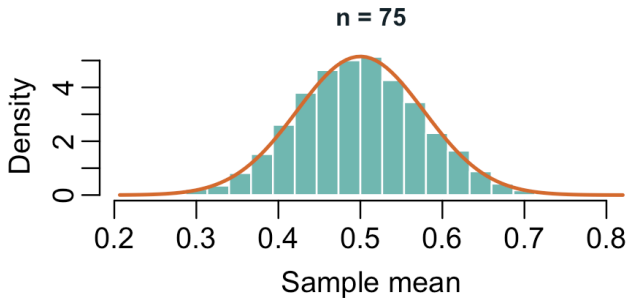
Why this n? Central shape improves over 30-60; 75-80 are borderline across seeds. At 90, modest right skew remains, but valid SE and larger neighbors support the compromise.

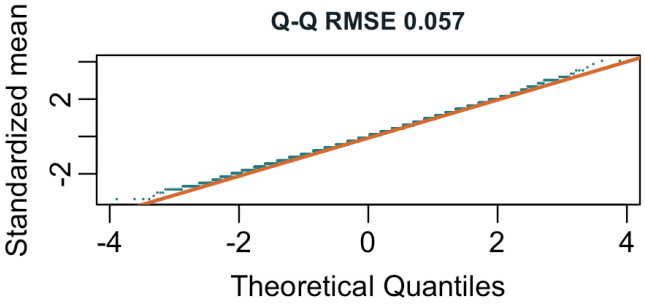
Observation: Correct mean/SE does not eliminate tail curvature.

Binomial small n = 75

iid Binomial(size=5,p=.10); mean .5, variance .45. Theory expectation: skew and lattice steps.

Tested: 1-1000 (29 values); neighborhood 60, 75, 80.





Metrics: skew 0.109; excess kurtosis 0.050; QQ 0.057.

Why this n? The bell and Q-Q line are useful apart from steps; skew is modest and SE agrees. At 60, QQ is about .067 in both seeds versus .057-.060 at 75; larger neighbors are comparable.

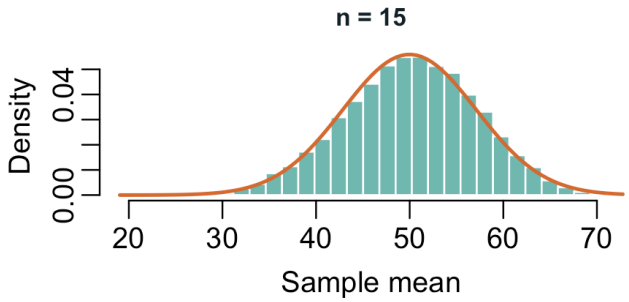
Observation: Binomial size=5 is NOT CLT sample size n.

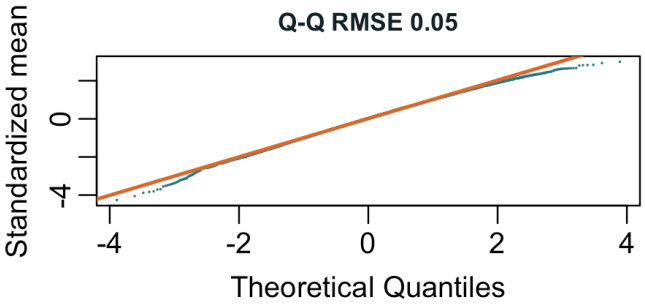
08 / Mixtures and varying probabilities

Synthetic ages / Extra credit — n = 15

iid synthetic ages 0-100; mean 49.981, SD 27.635. Theory expectation: bounded mixture smooths.

Tested: 1-500 (27 values); neighborhood 12, 15, 16.





Metrics: skew –0.191; excess kurtosis –0.039; QQ 0.050.

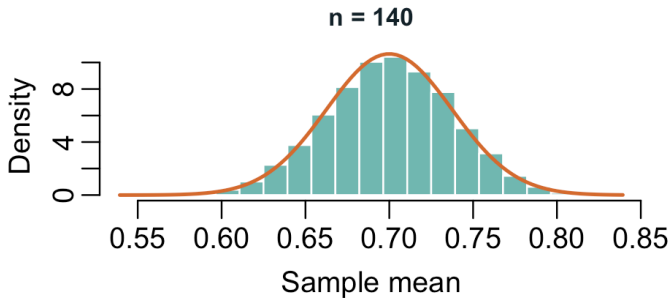
Why this n? The mixture has become a bell with modest left curvature and correct SE. At 5-10 asymmetry is clearer; 12 is borderline. Nearby 16 and larger values support 15.

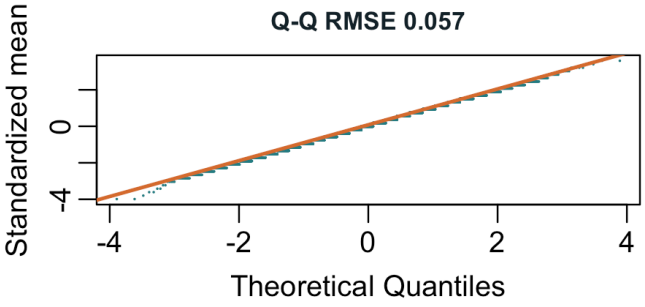
Observation: The irregular mixture is faster than the Exponential.

Basketball shots / Extra credit — n = 140

Independent shots, p_i=.90 to .50; not identical. Theory expectation: steps diminish with n.

Tested: 1-1000 (29 values); neighborhood 125, 140, 150.





Metrics: skew –0.052; excess kurtosis –0.039; QQ 0.057.

Why this n? The bell, small skew and fine steps support 140 with the correct varying-p SE. At 110-125 gaps are clearer; nearby 150 is comparable under both seeds.

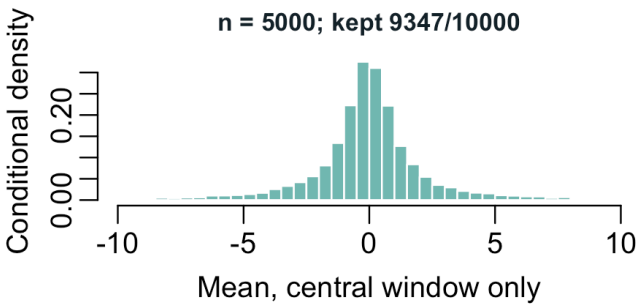
Observation: Independent does not mean identically distributed.

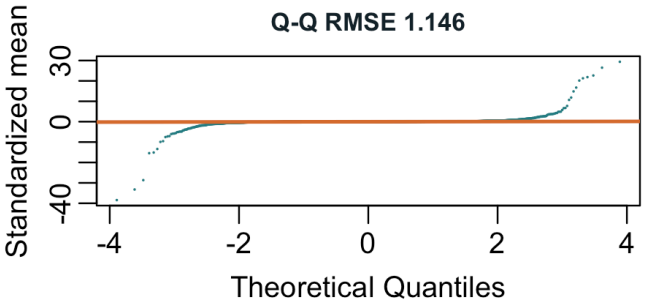
09 / Assumption stress tests

Cauchy — No finite n

iid Cauchy(0,1); no mean or variance. Theory expectation: averages never approach Normal.

Tested: 1-5000 (26 values); neighborhood 1, 100, 5000.





At n=5000, descriptive only: skew –3.468; excess kurtosis 649.912; QQ 1.146.

Histogram: [–10,10] window, conditional density; kept count labeled. Q-Q and metrics use all B means.

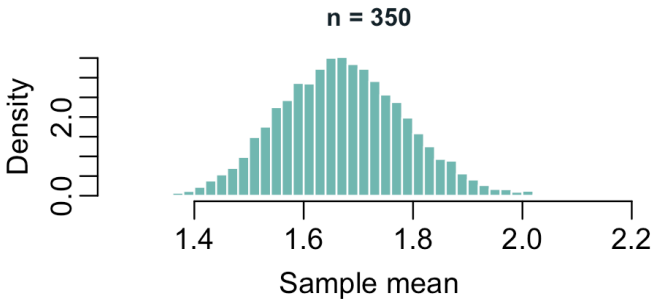
Why this n? At 5000, Q-Q departures remain extreme; a central bell does not rescue the tails. Theory proves averages stay Cauchy for every n, so no finite choice is assigned.

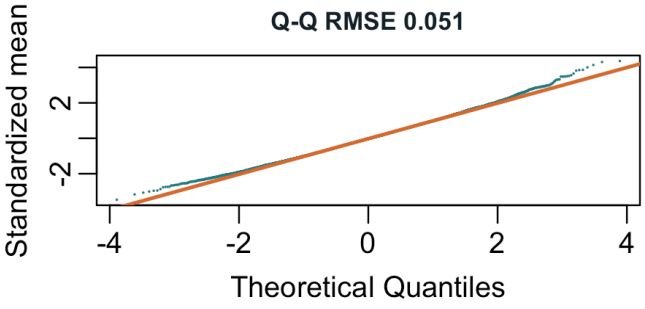
Observation: Finite-vector moments have no population targets.

Dependent failure / Extra credit — n = 350

Next rate=exp(1.25*(previous time-1)); first rate=1. Theory expectation: covariance matters.

Tested: 1-1000 (29 values); neighborhood 300, 350, 400.





Metrics: skew 0.208; excess kurtosis 0.097; QQ 0.051.

Why this n? At 350 the empirical bell and Q-Q line have modest right skew and agree across seeds. At 100-200 bends are clearer; 300 is borderline. This does not establish a dependent CLT.

Observation: No iid SE; asymptotic justification remains unresolved.

10 / Why not a smaller n?

Actual tested neighbors are compared below, not hypothetical failures between them. QQ is primary-seed Q-Q RMSE. Full graphical progression stays in the separate investigations.

Case	Lower n: QQ	Choice n: QQ	Higher n: QQ	Reservation
Normal	None below 1	1: 0.011	2: 0.012	Exact at 1
Six-sided die	9: 0.062	10: 0.057	12: 0.052	9 is borderline
Exponential	80: 0.063	90: 0.053	100: 0.031	80 varies across seeds
Binomial small	60: 0.067	75: 0.057	80: 0.058	60 has coarser steps
Dependent failure	300: 0.059	350: 0.051	400: 0.043	300 has more skew
Basketball shots	125: 0.067	140: 0.057	150: 0.058	125 has clearer gaps
Synthetic ages	12: 0.060	15: 0.050	16: 0.052	12 bends more left

Improvement is not a sudden switch

Die 9 and 10 are close, as are age 12 and 15. I do not claim every smaller integer fails: some were not tested and neighboring choices may be defensible under a looser tolerance. The selected value is a first convincing practical compromise within this grid.

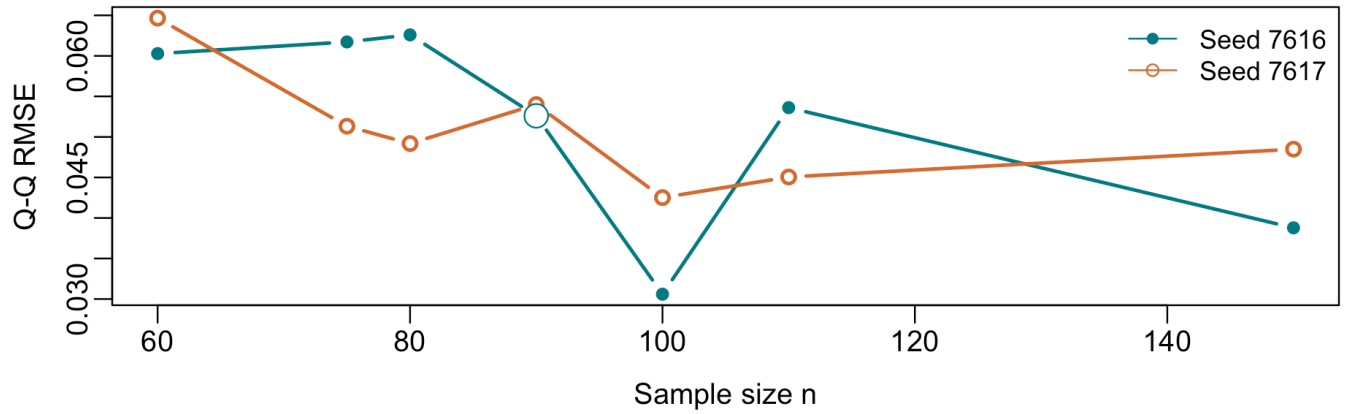


Figure 2: Exponential local search: both fixed seeds fluctuate around broadly improving shape. The highlighted 90 is a judgment, not a numerical-threshold crossing.

Why it matters: a larger n can have a worse finite-B diagnostic than its neighbor. The prettiest plot or first low QQ score would overstate boundary precision. Concordant visual evidence, valid theory and stability support the choice; uncertainty remains.

11 / Where the CLT became complicated

Case	What happened	Why / conclusion
Cauchy	Extreme Q-Q departures remain at 5000	Averages stay Cauchy; missing moments make this impossible, not slow
Dependent failure	Empirical means form a bell near 350	Covariance changes variance; empirical approximation, no proved dependent CLT
Basketball shots	Fine steps remain near 140 despite small skew	Independence survives, identical distribution does not; sum individual variances
Synthetic age mixture	Irregular population smooths by 15	Bounded iid observations; complexity alone does not imply slow convergence

Heavy tails: apparent stability can mislead

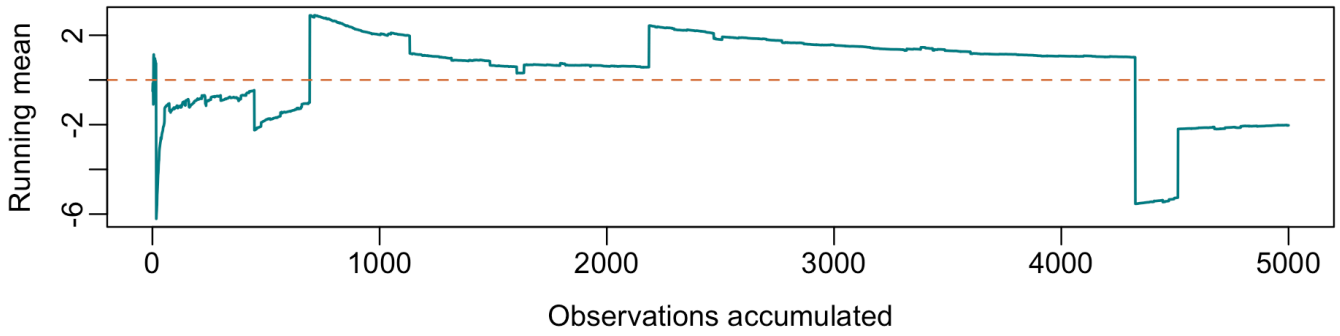


Figure 3: One Cauchy path, seed 7616. Jumps persist through 5,000 observations. Zero marks location, not population mean; one path is not a sampling distribution.

Cauchy's characteristic function is exp(-abs(t)); independence gives exp(-abs(t)/n)^n = exp(-abs(t)) for its average. It keeps the same law at every n. Simulation illustrates this theory; a finite search alone would not prove impossibility.

Dependence and mixtures: different complications

Long operating times raise the next failure rate, creating a compensating pattern. Individual second moments are bounded because rates are at least exp(-1.25), but a dependent CLT is not established. Covariance and the initial condition prevent borrowing the iid SE.

The synthetic age model stays iid and bounded: weights .045 exp(-age/5) + .030 exp(-.5((age-65)/10)^2), normalized on 0-100. Fast smoothing contrasts with the right-skewed Exponential; it is **not** a historical mortality claim or a result about every mixture.

12 / Validation: moments and a second seed

SE is the SD of the B simulated means, not individual observations. Agreement with theory checks center/spread; it does not establish a Normal shape.

Case	n	Mean sim.	Mean theory	SE sim.	SE theory	SE error %
Normal	1	-0.0013	0.0000	1.0012	1.0000	0.1230
Six-sided die	10	3.4947	3.5000	0.5394	0.5401	0.1213
Exponential	90	1.0017	1.0000	0.1064	0.1054	0.8988
Binomial small	75	0.4996	0.5000	0.0774	0.0775	0.0411
Cauchy	None	N/A	N/A	N/A	N/A	N/A
Dependent failure	350	1.6716	N/A	0.1155	N/A	N/A
Basketball shots	140	0.7004	0.7000	0.0377	0.0375	0.6188
Synthetic ages	15	50.0519	49.9810	7.1281	7.1352	0.0998

Appropriate references: iid finite-variance cases use SE = sigma/sqrt(n). Basketball uses mean(p_i) and Var(Xbar) = sum[p_i(1-p_i)]/n^2; its mean is .70 for n>=2, .90 for n=1.

At basketball n=140, correct SE is 0.03746, versus 0.03873 from the incorrect identical-p shortcut. Cauchy has no population moment targets. The dependent process has no supplied analytic mean/SE; baseline mean 1 is not the full-process target.

Exponential boundary: seeds 7616 and 7617

The alternate run repeats all 215 case/n combinations at B = 10,000. These are actual boundary rows, not a search for favorable seeds.

n	Seed	Mean	SE	Skew	Kurt.	QQ
80	7616	1.0017	0.1126	0.2646	0.1248	0.0626
80	7617	0.9991	0.1104	0.2049	0.0892	0.0492
90	7616	1.0017	0.1064	0.2144	0.0319	0.0526
90	7617	0.9992	0.1054	0.2240	0.0908	0.0540
100	7616	1.0002	0.1000	0.1194	0.0290	0.0306
100	7617	0.9995	0.1005	0.1724	-0.0043	0.0425

Seed conclusion: exact diagnostics change, but n=90 remains a defensible central approximation under both seeds. The greater discrepancy at 80 supports its borderline reservation. The same working choices remain practical candidates; an automatic algorithm did not independently rediscover every integer.

Reproducibility: seed directories are separate, and saved vectors reproduce CSV B, mean, SD and QQ. The October 6 fresh-directory full rerun matched generated tables byte-for-byte on the same R version. This revision rechecks saved evidence, not a new full simulation. Detailed investigations retain the all-population seed comparisons.

13 / Conclusions and the evidence archive

1. **No universal n=30:** practical choices span 1-350; Cauchy has no finite choice.
2. **Shape matters, complexity alone does not:** the bounded age mixture smooths faster than the right-skewed Exponential.
3. **Discreteness is not automatic failure:** useful die, Binomial and basketball approximations can retain Q-Q steps.
4. **Moment agreement is not shape agreement:** mean/SE checks must accompany graphical and shape evidence.
5. **Assumptions change the answer:** Cauchy lacks moments, machine times require covariance, and shots require individual probabilities.
6. **A boundary needs an argument:** nearby evidence, theory and two seeds matter more than one pretty plot or isolated score. Borderline minima remain uncertain.

Detailed analysis-file map

Paths are relative to github.com/babaJaan01/Monte_Carlo_CLT. Each `.qmd` contains population/theory, tested progression, plots, justification, limitations and seed comparison. Rendered HTML has matching basenames in `docs/analyses/`.

Population	Detailed analysis file
Normal	analyses/01_standard_normal.qmd
Six-sided die	analyses/02_six_sided_die.qmd
Exponential	analyses/03_exponential.qmd
Binomial small	analyses/04_binomial_small.qmd
Cauchy	analyses/05_cauchy.qmd
Dependent failure	analyses/06_extra_dependent_failure.qmd
Basketball shots	analyses/07_extra_nonidentical_bernoulli.qmd
Synthetic ages	analyses/08_extra_age_at_death.qmd

Results / plots: primary, sensitivity and final-summary CSVs in `results/`; judgments in `decisions.csv`; summary visuals in `figures/`; individual plots in `docs/analyses/`. Mean vectors are regenerated, not tracked in Git.

Reproduce: restore `renv.lock`, run `scripts/run_project.R`, `scripts/validate_results.R`, `scripts/render_summary.R`, then `quarto render`. Statistics use base R; knitr/rmarkdown support rendering; Quarto supplies Typst. Exact commands are in README.

Sources: original guidance, Class 05 CLT and Class 06 distributions/project lessons, plus all three instructor models, remain unchanged in `references/`. Updated guidance defines five required and three extra cases. Expectations are retrospective; no undocumented personal prediction or surprise is manufactured.